



Research Methods in Financial Engineering

Graduate School of Finance
KAIST College of Business

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What Will be Covered in the Course?

- I will try to review the basic foundation of financial engineering, and review the empirical methods and asset pricing literature to do some 'research' in financial engineering.
 - Binomial tree models and risk-neutral pricing
 - Consumption-based asset pricing models and GMM
 - Factor pricing models and asset pricing tests
 - Anomalies

Class Organization

- Research Methods in Financial Engineering I & II
 - Every class starts at 8:30, and ends at 10:15. (105 minutes, instead of 75 minutes)
 - Research Methods in Financial Engineering I starts on September 1 and ends on October 15. (7 weeks)
 - Research Methods in Financial Engineering II starts on October 20 and ends on November 19. (5 weeks, not 8 weeks)

Class Organization (2)

- Research Methods I
 - Final exam: October 15 (Wednesday)
- Research Methods II
 - No exam
 - Students will submit & present their drafts of Master's thesis and will be evaluated by the quality of the drafts as well as their presentations.

Picking up a Thesis Topic

- First, you need to ask yourself what area you are interested in.
 - Since you are writing a thesis for the financial engineering Master's degree, the topic should be closely related to the finance area.
- If the broad area for the thesis is decided, then you need to look over the existing literature in the area, and how the other people have thought of the issues in the area.
- Next, you need to fix a concrete question you want to deliver in your thesis.
- Find a key paper (or papers) for the question, and then replicate the paper with your own data. Then think about how to adjust/expand the key paper to the one that suits well with your idea or data.

Thesis Development and Presentations

- The defense for Master's thesis is expected to be held early December.
- You should submit a three-page thesis proposal by September 8, 2025.
 - It should contain:
 - Title
 - Motivation
 - Key paper(s)
 - Hypotheses and their reasoning
 - Data
 - References

Thesis Development and Presentations (2)

- You should submit a more refined seven-page proposal by October 1, 2025.
 - It should contain:
 - Title
 - Motivation
 - Key paper(s)
 - Hypotheses and their reasoning
 - Data
 - Key empirical results
 - References

Thesis Development and Presentations (3)

- The presentation schedule will be fixed by mid-October.
 - October 20 (Mon), 22 (Wed), 27 (Mon), 29 (Wed), and November 3 (Mon), 5 (Wed), 10 (Mon), 12 (Wed), 17 (Mon), 19 (Wed)
 - Four (or five) students are expected to present their drafts for each class.

TA Information

- If you have any questions, you can ask me or TA for this class.
- Teaching Assistance
 - Jiwon Eo(littleeo@kaist.ac.kr)
 - Office: S391
- My Office: S390 (Phone: 958-3521, jkkang@kaist.ac.kr)